Embodied 3D Spatial Perception Through Layered Robotic Skin

Self-Supervised Representation Learning from Autonomous Motion

Anonymous Author(s)

Independent Research

correspondence: [withheld for review]

Abstract

We propose a novel paradigm for endowing robots with rich three-dimensional spatial perception through a multi-layered tactile skin architecture coupled with self-supervised learning from autonomous motion. Unlike conventional approaches that treat tactile sensing as a passive input channel for contact detection or force measurement, our framework conceptualizes the robot's body as an active perceptual apparatus: the layered skin generates high-dimensional spatiotemporal pressure signals across multiple depth planes, while the robot's own motion provides a self-generated supervisory signal that links tactile patterns to kinematic states. We argue that this embodied self-supervision enables the emergence of a 3D spatial representation without explicit human annotation, as the robot learns to predict tactile consequences of its actions and, conversely, infer environmental geometry from contact patterns. We detail the proposed architecture—comprising a biomimetic three-layer skin, a sensor-aware temporal encoder, and a contrastive predictive coding objective—and outline a research program including simulation-to-real transfer, large-scale self-collection protocols, and downstream task evaluation. We position this work as a foundational step toward robots that *feel* their world in three dimensions.

Keywords: robotic tactile perception, multi-layer electronic skin, self-supervised learning, embodied intelligence, 3D spatial representation, human-robot interaction

I. Introduction

The dominant paradigm in robotic perception is vision-centric. Cameras provide rich semantic information at a distance, but they fail catastrophically under occlusion, low light, translucency, and specular reflection—precisely the conditions in which physical contact becomes necessary. Yet when a robot touches an object, it receives a deluge of information that vision cannot: local geometry, compliance, texture, thermal properties, and the temporal evolution of contact. This information is inherently three-dimensional, distributed across the body, and temporally structured by motion.

Biological organisms solve this through skin—a multi-layer mechanosensory organ that transduces deformation into electrical signals across multiple depth planes. Human skin contains four principal mechanoreceptor types (Merkel cells, Meissner corpuscles, Ruffini endings, and Pacinian corpuscles), each tuned to different spatial scales, depths, and temporal frequencies. Crucially, the spatial structure of the body and the patterns of self-generated motion allow the brain to learn which tactile signals correspond to external objects versus self-touch—a distinction that emerges through sensorimotor experience rather than being hard-coded.

We propose to import this principle into robotics: **equip robots with a multi-layer pressure-sensitive skin and let them learn 3D spatial perception through self-generated motion, without human annotation.**

> **Core hypothesis:** A robot equipped with multi-layer distributed tactile skin, when allowed to move autonomously in an unstructured environment, can learn a 3D spatial representation of its surroundings entirely through self-supervision—the robot's kinematic state provides the supervisory signal, and the multi-layer tactile response provides the rich observation from which geometry, material, and contact dynamics are inferred.

II. Related Work

A. Electronic Skin and Distributed Tactile Sensing

Electronic skin (e-skin) research has produced remarkable sensors: capacitive, piezoresistive, piezoelectric, and optical tactile sensors can measure normal/shear forces, vibration, temperature, and humidity at high spatial resolution. Recent work

includes 3D-stacked architectures (Kim et al., Nano-Micro Letters 2024) that achieve simultaneous pressure-temperature perception at 92.6 sensors/cm²; biomimetic layered skins (Boutry et al., Sci. Robot. 2018) that decouple normal and shear forces; and 3D-printed conformal tactile arrays (Liu et al., 2024) enabling complex-curved surface coverage. Tesla's recent patent (US20260310299A1) discloses a 3D flexible tactile array for the Optimus Gen-3 hand, covering all phalanges and palm.

However, these works largely treat the sensor as a hardware endpoint: data is collected, calibrated, and fed into task-specific models. None has proposed a system where the full-body multi-layer skin output becomes the training signal for a general-purpose 3D spatial representation learned through self-motion.

B. Self-Supervised Tactile Representation Learning

Recent work in self-supervised tactile representation has made significant strides. Sparsh (Higuera et al., 2024) learns general-purpose tactile encoders from 460,000+ tactile images using masked modeling and self-distillation, demonstrating transfer across sensors and tasks. CTTP (Rodriguez et al., ICRA 2025) uses contrastive learning to bridge heterogeneous tactile sensors. Sparsh-Skin (Sharma et al., 2025) extends this to magnetic tactile skin distributed across dexterous hands, conditioning representations on sensor positions. T3 (2024) trains a multi-sensor multi-task model on 3M+ samples across 11 tasks.

Our proposal differs in scope and mechanism: (1) *multi-layer* skin provides depth-resolved information unavailable to single-plane sensors; (2) the robot's *entire body* is the sensing surface, not just fingertips; (3) the supervisory signal is the robot's own *kinematic state during autonomous exploration*, not paired cross-sensor data or visual supervision.

C. Embodied Learning and Sensorimotor Contingency

The idea that perception emerges from action has deep roots in developmental psychology (Piaget) and neuroscience (Helmholtz's "unconscious inference," von Holst & Mittelstaedt's reafference principle). In robotics, active perception and sensorimotor learning have been explored in limited domains. However, the specific proposal of using full-body multi-layer tactile self-supervision to learn 3D spatial representations—a "tactile world model"—remains unexplored.

III. Proposed Architecture

A. Multi-Layer Skin Design

We propose a three-layer biomimetic skin inspired by the stratified structure of human skin:

Table 1: Proposed three-layer skin architecture

Layer	Depth	Sensing modality	Primary signal
L1 (Epidermal)	Surface	High-density capacitive pressure	Rapid transient contact, texture, slip
L2 (Dermal)	Mid-depth	Piezoresistive force + shear	Sustained pressure distribution tangential forces
L3 (Subcutaneous)	Deep	Strain + deformation field	Whole-body deformation proprioceptiv coupling

The key insight is that each layer captures a different spatiotemporal scale. L1 responds to high-frequency transients (texture, incipient slip); L2 encodes sustained contact geometry; L3 captures low-frequency deformation that propagates through the body. The *combination* of signals across layers provides depth-resolved contact information—a 3D "image" of the contact event rather than a 2D pressure map.

B. Sensor-Aware Temporal Encoder

Raw skin data consists of thousands of channels across hundreds of patches distributed over the body surface. We propose a hierarchical encoder:

1. Patch-level encoding: A small temporal convolutional network (TCN) processes the multi-channel signal from each skin patch independently, producing a patch embedding that is conditioned on the known 3D location of that patch on the robot body.

2. Cross-patch attention: A transformer-style attention mechanism aggregates patch embeddings across the body, allowing spatially distant contacts to be related (e.g., an object contacting both the fingertip and the palm simultaneously).

3. Temporal fusion: A recurrent or state-space model integrates the cross-patch representation over time, producing a latent state that evolves with the robot's motion.

$$z_t = f_\theta(T_{t-K:t}^{(1)}, T_{t-K:t}^{(2)}, T_{t-K:t}^{(3)}, q_{t-K:t}, \dot{e}_{t-K:t}) \quad (1)$$

where $T^{(i)}$ denotes the signal from layer i, q denotes joint configuration, and $\dot{e}$ denotes joint velocity. The encoder is sensor-position-aware: each patch embedding is augmented with its known body-frame coordinates.

C. Self-Supervised Objective

We formulate the learning problem as a combination of three complementary self-supervised objectives:

1. Contrastive Predictive Coding (CPC): Given the current tactile latent z_t and the current kinematic state $s_t = (q_t, \dot{e}_t)$, predict the tactile latent at a future timestep z_{t+k} under the same action sequence. Positive pairs are temporally adjacent samples from the same trajectory; negatives are sampled from different trajectories.

$$L_{CPC} = -\mathbb{E}[\log (\exp(f(z_t, s_t) \cdot W z_{t+k}) / \Sigma_{z'} \exp(f(z_t, s_t) \cdot W z'))] \quad (2)$$

2. Kinematic-Tactile Alignment: Train the tactile encoder such that the latent representation, when combined with the kinematic state, can predict the next kinematic state—and conversely, the kinematic state combined with the current tactile latent can predict the next tactile state. This bidirectional prediction forces the model to discover the causal relationship between action and tactile consequence.

$$L_{align} = \| g(z_t, s_t) - s_{t+1} \|^2 + \| h(s_t, z_t) - z_{t+1} \|^2 \quad (3)$$

3. Cross-Layer Consistency: Different layers should agree on the global contact event while preserving their distinct local information. We use a mutual information maximization objective between layer-specific representations, tempered by a redundancy reduction term.

$$L_{cross} = \Sigma_{i \neq j} [I(z^{(i)}_t; z^{(j)}_t) - \lambda \cdot R(z^{(i)}_t, z^{(j)}_t)] \quad (4)$$

The total objective is a weighted sum: $L = \alpha L_{CPC} + \beta L_{align} + \gamma L_{cross}$.

IV. Self-Data Collection Protocol

A. Autonomous Exploration

Data collection requires no human annotation. The robot is placed in an environment with diverse objects and surfaces, and executes a randomized but safe exploration policy:

- **Free-space motion:** The robot moves all joints through their range, generating baseline self-motion signals (proprioceptive calibration).
- **Surface contact:** The robot approaches surfaces at varying angles, speeds, and forces, producing contact onset signals across all skin layers.
- **Object interaction:** The robot grasps, pushes, rolls, and slides objects of varying geometry and compliance.
- **Whole-body contact:** The robot brushes against walls, crawls on textured floors, and navigates cluttered spaces—activating skin across the entire body surface.

Every frame records: (a) full-layer skin data $T_t^{(1,2,3)}$, (b) kinematic state $s_t = (q, \dot{e}, \ddot{e})$, (c) IMU/accelerometer data, and (d) optional synchronized camera data (for bootstrapping only, later removed).

B. Curriculum and Scaling

We propose a three-stage curriculum:

Stage 1 (Kinematic grounding): Free-space motion establishes the baseline mapping between self-motion and skin deformation. This is the "I am moving" phase, analogous to infant body schema development.

Stage 2 (Contact discovery): Introduction of surfaces and simple objects. The robot learns to distinguish self-generated deformation from externally-induced deformation—the reafference principle implemented computationally.

Stage 3 (Rich interaction): Complex manipulation, whole-body contact, and multi-object scenarios. The representation generalizes to novel geometries and materials.

> **Key advantage:** Data is generated autonomously at near-zero marginal cost. A robot operating continuously can produce millions of frames per week. The scale of self-collected tactile interaction data could dwarf all existing tactile datasets combined.

V. Learning 3D Spatial Representation

A. From Contact Patterns to Geometry

The multi-layer skin, when deformed by contact with a 3D object, produces a characteristic signal profile across layers. The surface layer (L1) captures the contact boundary; the mid-layer (L2) captures the pressure distribution; the deep layer (L3) captures the deformation field propagating through the body. Together, these form an overcomplete representation from which the 3D geometry of the contact surface can be reconstructed.

Consider a robot fingertip pressing against a curved object. The L1 signal localizes the contact region. The L2 signal encodes the pressure gradient across that

region. The L3 signal captures how the force propagates into the finger structure. By correlating these signals with the known kinematic state, the model learns to infer the curvature, orientation, and compliance of the object surface at the contact point.

B. Temporal Integration and Predictive Models

As the robot moves, it "paints" a 3D map of its environment through sequential contact. Each contact event adds a local 3D patch to the growing spatial representation. The temporal encoder maintains a persistent latent state that accumulates these patches into a coherent model of the surrounding space.

This is fundamentally different from vision: the robot does not infer geometry from a 2D projection; it directly measures the 3D geometry through physical interaction. The spatial representation is metric, not perspective-dependent.

C. Emergent Capabilities

We hypothesize that the following capabilities will emerge from the proposed training paradigm:

- **Self-other distinction:** The model learns to classify tactile signals as self-generated vs. externally-caused, enabling detection of unexpected contacts.
- **Material inference:** The temporal-spectral signature across layers encodes compliance, friction, and texture.
- **Contact localization in 3D:** Multi-patch cross-attention resolves the 3D position and orientation of contact events.
- **Predictive touch:** The model anticipates tactile consequences of planned actions, enabling proactive control.
- **Transfer across body parts:** The sensor-position-aware encoder generalizes learned representations across different body regions.

VI. Experimental Validation Roadmap

A. Simulation Phase

We propose a high-fidelity soft-body simulation (e.g., Isaac Sim with soft-contact models) to validate the architecture before hardware deployment. The simulator should model:

- Multi-layer deformation mechanics
- Distributed pressure sensor noise and drift
- Diverse object geometries and material properties
- Robot kinematics and self-collision

Metrics: (1) reconstruction error of object geometry from tactile-only input; (2) classification accuracy of material properties; (3) contact localization error in 3D; (4) transfer performance to unseen objects.

B. Hardware Prototype

A minimal viable prototype consists of a 3-DoF robotic finger or palm segment equipped with the three-layer skin. The prototype should demonstrate:

- Real-time signal acquisition from all three layers simultaneously
- Robustness to cyclic loading (durability > 10^5 cycles)
- Cross-talk suppression between adjacent patches
- Integration with the proposed encoder architecture

C. Full-Body Deployment

Scale to a humanoid torso or arm with >1000 skin patches. Evaluate on:

- Whole-body obstacle detection without vision
- Blind manipulation of novel objects
- Human-robot contact safety (pressure distribution > force threshold)
- Long-horizon spatial memory formation

VII. Discussion

A. Comparison with Visual Perception

Visual perception is fundamentally indirect: it infers 3D from 2D projections, relying on priors, stereopsis, or active camera motion. Tactile perception via multi-layer skin is direct: each layer samples a different depth plane of the contact event, and motion sweeps this multi-plane sensor across the object surface. We argue that for near-field spatial understanding—the regime in which robots must physically operate—layered tactile sensing is not merely complementary to vision but potentially superior.

B. The Role of AI in Sensor Data Fusion

The central role of AI in this framework is not to "process sensor data" in the conventional sense. Rather, AI *discovers the mapping* between multi-layer tactile signals and 3D spatial structure through interaction. The robot learns what its skin "means" by moving and observing the consequences. This is a radical departure from engineered feature extraction: the representation is learned, not designed.

This approach also suggests a new interpretation of embodiment: the robot's body is not just an actuator attached to a sensor suite; it is a self-calibrating perceptual instrument whose geometry, material properties, and sensor layout are all known by the learning system and exploited for spatial inference.

C. Limitations and Future Directions

Several challenges remain: (1) Sensor drift and hysteresis in soft materials may corrupt the self-supervisory signal; (2) The combinatorial space of contact modes may require impractical amounts of data; (3) Real-to-sim gap in soft-contact physics; (4) Computational cost of processing thousands of high-frequency channels in real-time. Future work should explore sparse attention mechanisms, event-driven sensing, and neuromorphic processing to address these constraints.

VIII. Conclusion

We have proposed a paradigm in which robots acquire 3D spatial perception through a multi-layer tactile skin and self-supervised learning from autonomous motion. The architecture—biomimetic three-layer skin, sensor-aware temporal encoder, and contrastive predictive objective—provides a concrete path toward robots that feel and understand their physical environment in three dimensions. The self-supervision protocol requires no human annotation: the robot's own motion and kinematic state provide the supervisory signal, while the multi-layer tactile response provides the rich observation. We believe this represents a fundamental shift in how robotic perception can be achieved: from passive sensing of external signals to active, embodied discovery of spatial structure through touch.

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